

OLIVER ATHERTON

Huddersfield, U.K. | 07956555742 | oliverhatherton@gmail.com | linkedin.com/in/oliverhatherton | github.com/oliverhatherton

EXPERIENCE

Software Engineer Intern

Sep 2025 — Sep 2026

Babcock International Group

Bristol, UK

- Selected 1 of 3 from 140 candidates (~top 2%) for competitive software engineering placement
- Restored production application offline for ~1 year to operational status in under 2 weeks by self-learning Rust/Tokio/Tauri and resolving critical backend and frontend defects
- Delivered end-to-end feature (UI, backend, DSL) in 4 days (~50% faster than estimate) by consolidating integration work
- Refactored multiple REST endpoints into a unified backend architecture, reducing duplication and improving maintainability while preserving system behaviour
- Achieved full productivity in 3 days on second project, contributing to sprint delivery by implementing backend API features in Spring Boot
- Built and extended internal tooling (SNMP simulator + Azure DevOps workflows), reducing setup and bug triage time for developers
- Worked within constrained hardware lab and security environments, adapting development and testing workflows to maintain delivery velocity

LEADERSHIP EXPERIENCE

Backend Lead

Oct 2025 — Oct 2025

Babcock Internal Hackathon

- Led backend development for convoy management platform, enabling ~90% faster report generation via single-click PDF export
- Delivered 42% of total commits (team of 4) while onboarding a junior developer and architecting and implementing backend services

PROJECTS

Distributed Task Queue System | RabbitMQ, Redis, Node.js, Docker, Kubernetes, Terraform

Feb 2026 — May 2026

- Built distributed task queue handling 3,200 req/sec and 500 concurrent users, validated with K6 load testing
- Designed microservices architecture (API, Worker, Scheduler) with asynchronous messaging, supporting horizontal scaling to 100+ workers
- Implemented fault tolerance using retries with exponential backoff, dead-letter queues, and lease-based job reassignment
- Added observability with Prometheus and Grafana dashboards for latency, throughput, and failure monitoring
- Implemented priority-based message routing and API-key authentication, ensuring secure and latency-optimised processing for time-sensitive workloads
- Designed worker heartbeat and lease-based coordination mechanism to detect failures and reassign jobs in real time
- Deployed production-ready system using Docker, Kubernetes, Terraform, and AWS, with load testing via K6 and public deployment on Render

SKILLS

Languages: Java, TypeScript, Python, Rust, SQL

Backend & Systems: Spring Boot, Node.js, REST APIs, Distributed Systems, Microservices, Express.js

Infrastructure: Docker, Kubernetes, Terraform, AWS, Azure, CI/CD

Data & Messaging: PostgreSQL, Redis, RabbitMQ, MongoDB

Testing: JUnit, Spring Test, Jest, Cypress

Computer Science Fundamentals: Operating Systems (91%), Software Design (81%), Computer Architecture (79%)

EDUCATION

University of Huddersfield, West Yorkshire

Sep 2023 — Jul 2027

BSc in Software Engineering

GPA: Expected First Class Honours